SG3 - What Affects Tennis Win Percentages

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Summary / Research Question

The goal of this project is to determine which match statistics are most strongly associated with winning in professional tennis and to evaluate whether match outcomes can be predicted using these statistics. Specifically, we ask: **Can we predict whether a player wins a match based on differences in performance metrics such as aces, double faults, and serve efficiency?**

To explore this, we analyze ATP match data from two different seasons (2014 and 2024), allowing us to capture patterns across time and ensure the robustness of our findings.

Introduction

Tennis is a sport where matches are often decided by small differences in performance, making it an interesting setting for data analysis. While fans and analysts often emphasize factors like player skill, experience, or physical attributes, it is not always clear which specific match statistics are most important in determining the outcome of a match. With the growing availability of detailed sports data, it is now possible to explore these questions using data driven methods.

The goal of this project is to identify which aspects of a tennis match are most strongly associated with winning and to determine whether match outcomes can be predicted using these statistics. In particular, we focus on performance based metrics such as aces, double faults, and serve efficiency, and compare their impact to player characteristics like age and height.

To answer this question, we analyze ATP match data from Grand Slam tournaments in 2014 and 2024. By comparing these two seasons, we are able to examine whether patterns remain consistent over time. Using engineered features that capture the differences between winners and losers, we apply machine learning models to evaluate how well match outcomes can be predicted and to better understand which factors contribute most to success in professional tennis.

Methodology

Data Preparation

We started by reading in the tennis data sets for 2014 and 2024, combining them, and making everything lowercase so that it was clear and easy to use. We dropped the variables that are not relevant to the analysis and narrowed the rows to ones that fell into the 4 major tournaments (Australian Open, Roland Garros, Wimbledon, and US Open).

Creating New Features

To look at meaningful differences between the winner and losers in the data set, we made variables of the differences, subtracting the value for the loser from the value of the winner. We did this with several variables including number of aces, number of double faults, breaked points saved, age, height, etc.

To be more precise, for each match (row) in the dataset, some of the variables (columns) had two copies - one for the winner and one for the loser of the match. We represent the winner variables with the vector W and we use L to denote the loser variables. For each of these variables with two copies (W_i for winner and L_i for loser), we compressed them into one variable M_i with the value $W_i - L_i$. Thus, we ended up with a vector of features $M = W - L$.

These new variables demonstrate the difference between the winners and losers. By finding the means of these new variables, we can find which ones had the biggest differences and therefore which ones are the best predictors of winning.

Plotting New Features

We created histograms with density curves for the following new variables: aces difference, double fault difference, 1st serve points won difference, and 2nd serve points won difference. The resulting plots show the mean differences. If they center at 0, then there is likely no difference. If they center a positive number, that means that the winners had higher values than the losers. If they center a negative number, then the losers had higher values for that variable than the winners.

We also created a heat map to look at the correlation between the new variables. That helps us determine if there is any correlation between the variables, answering the question: “When one advantage increases, does another advantage also increase?”. It can show us underlying patterns in the data as well as possible redundancies. From the results of the heatmap, we determined that none of the variables are too strongly correlated to where we would have to remove any.

Model

Before training a model, we had to create a set of samples for training/testing the model. To do this, we started with the dataset, but limited the features to the difference features which we engineered. So our feature vector for each sample was the previously described M = W - L. We labeled all such samples with “winner”. We wanted some samples to label as “loser”, though, so we augmented the samples with new samples which were essentially negative versions of all the winner samples. They looked like L - W instead, and were labeled with “loser”. We converted the labels to appropriate numerical values based on the type of model we decided to use. Finally, we used 80% of the samples as training samples and the remaining 20% as testing samples.

We trained several types of models on the training samples, including logistic regression, random forest, SVM with standard scaling, and KNN.

Results

The performance of the best model in each category was evaluated using accuracy on the test set as the evaluation metric.

- Logistic Regression
    - 91.9% test accuracy
- Random Forest
    - 86.7% test accuracy
- SVM
    - 88.6% test accuracy
- KNN
    - 62.7% test accuracy

Logistic Regression achieved the highest accuracy, closely followed by Random Forest. These results indicate that the engineered features are highly informative and that match outcomes can be predicted with relatively high accuracy using performance statistics alone.

This follows closely with standard knowledge of men’s pro tennis. The in-match strategy has always been considered the deciding factor in matches. Serve mastery has decided matches since the beginning of tennis history, and only becomes more important as the games evolve and serves become faster, harder, and better placed.

This gives us an insight into tennis: anyone can be a tennis star if they master the art of strategy. It might be interesting to see if women’s pro tennis yields different results, as it is widely considered less serve-dominated than the men’s game. Perhaps mixing the data

together would give an average tennis player an idea of where to sharpen up their skills to bring their game to the next level.

Discussion

The strong performance of Logistic Regression suggests that the relationship between match statistics and winning is largely **linear and interpretable**. This is important because it allows us to understand which features contribute most to success. The SVM performed similarly well to logistic regression, likely for similar reasons. It only fell short of the logistic regression by a few percentage points.

Lower performance from KNN indicates that simpler distance-based or margin-based approaches may not capture the structure of the data as effectively as regression-based models.

Key insights from the analysis include:

- Higher ace differences are associated with a greater likelihood of winning
- Fewer double faults improve chances of winning
- Strong performance on both first and second serves is critical
- Break point performance plays an important role in match outcomes
- The fact that Random Forest performed similarly well suggests that there may also be **nonlinear interactions** between features, although these are not strictly necessary for accurate prediction.

Conclusion

Our analysis shows that serve performance is the most important factor in determining match outcomes in professional tennis. Players who perform better on first and second serves, hit more aces, and manage break points effectively are much more likely to win. In contrast, age and height have little impact.

Among the models tested, logistic regression performed the best, followed by SVM and random forest, showing that match outcomes can be predicted with high accuracy using performance statistics alone. Overall, these results highlight the importance of in match performance, especially serving, in achieving success at the highest level of tennis.

Links

Project GitHub Repository: https://github.com/wrigh854/cmse202-s26-project